

Senthil Nagappan

Forward-Deployed AI Experience & Human Systems Integration Leader

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PROFESSIONAL SUMMARY

Forward-deployed AI experience and human-systems leader with 18+ years embedding with users, engineering, product, risk, and client teams to move regulated systems from ambiguity to production. Pairs AI governance, accessibility, design systems, human factors, and production front-end engineering to make AI safe, usable, accountable, and measurable. 7+ years in compliance, risk management, and advisory delivery; directed a \$130M federal portfolio and scaled a cross-functional team to 15+.

FORWARD-DEPLOYED OPERATING MODEL

Embed in real workflows · **Frame** user, technical, and risk constraints · **Prototype** at production fidelity · **Deploy** inside the release cadence · **Validate** adoption, override, correction, and audit evidence.

CORE CAPABILITIES

Forward-deployed AI delivery · Field immersion & workflow discovery · Cross-functional problem framing · Embedded engineering partnership · Production-fidelity prototyping · Deployment instrumentation · Client & executive delivery · AI governance & model risk · NIST AI RMF · EU AI Act · ISO/IEC 42001 · Human-AI interaction · Explainability & trust UX · Human-in-the-loop oversight · WCAG 2.2 AA/AAA · Section 508 · React/TypeScript · Design systems · IEC 62366 · MIL-STD-1472H

EXPERIENCE

AI Experience Designer & HSI Lead

GE HealthCare (Brain Pulse Inc) · Mar 2025 – Present · Remote

- Work forward-deployed with product and engineering teams, translating field context, clinical risk, and governance requirements into release-ready product behavior.
- Design agentic clinical workflows where the model recommends and the clinician decides, with oversight tiers, override paths, source grounding, and known-limits disclosure.
- Prototype in React and TypeScript; ship accessible design-system components adopted across 10+ teams and map behavior to NIST AI RMF and IEC 62366 evidence.

Sr. UX & Accessibility Lead

Social Security Administration (Leidos) · Oct 2022 – Feb 2025 · Remote

- Worked forward-deployed inside federal program delivery, connecting caseworker needs, engineering constraints, accessibility policy, and 40+ monthly releases.
- Built and governed an enterprise design system across 15+ modules serving 50M+ citizens; traced 65% of interaction issues to 8 reused components, saving an estimated \$1.5M+ annually.
- Embedded WCAG 2.2 AA / Section 508 gates in CI/CD and advised leadership across a \$130M program portfolio; scaled and mentored a team to 15+.

Human Factors Engineer · Sr. UX & Inclusive Design Lead

Best Buy Health · Sep 2021 – Sep 2022 · Remote

- Embedded field research and use-related risk analysis into product, engineering, and validation decisions for connected medical-device experiences reaching 1M+ users.
- Ran summative usability validation supporting 510(k) human-factors evidence under FDA HFE, IEC 62366, and ISO 14971; contributed accessible BRIX design-system patterns.

Sr. UX & Program Lead

HHS / SAMHSA (Eagle Technologies) · Mar 2014 – May 2021 · Remote

- Led a forward-deployed client-delivery model spanning discovery, roadmap, stakeholder alignment, team growth, and release governance for national behavioral-health data services supporting 2M+ users.

Earlier Roles

AAMC · CareFirst BCBS · JPMorgan Chase · Morgan Stanley · 2007 – 2014

- Enterprise UX, accessibility, front-end delivery, and financial-services work that established the regulated-delivery and model-risk discipline now applied to AI.

SELECTED AI & PROTOTYPING WORK

Built a full-stack AI visual-assistance PWA for blind and low-vision users with scene description, OCR, and product identification; created an AI-experience pattern library spanning explainability, trust calibration, agentic hand-off, and human oversight; designed governance concepts for model inventory, evidence, red-team evaluation, and source-grounded AI.

EDUCATION & CREDENTIALS

B.E., Electronics & Communication Engineering (HCI / human-factors focus), Madurai Kamaraj University, 1999.

AIGP — AI Governance Professional (IAPP), in progress · IAAP CPACC + WAS (CPWA) · CUA · CSM · Salesforce Certified UX Designer, in progress · DAU HSI, in progress

STANDARDS & LEADERSHIP

WCAG 2.2 AA/AAA · Section 508 · FDA HFE / IEC 62366 · ISO 14971 · MIL-STD-1472H · MBSE/SysML · DoDI 5000.95 · NIST AI RMF · EU AI Act · ISO/IEC 42001 · SR 11-7

Vice President, Texas Astronomical Society; Cubmaster & Assistant Scoutmaster, Scouting America; Lead Coach, FIRST LEGO League.